

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	The seeds would be destroyed in the process so they would have none to sell	B1	2.4
		(1)	
(b)	$[S = \text{no. of seeds out of 24 that germinate, } S \sim B(24, 0.55)]$		
	$T = \text{no. of trays with at least 15 germinating. } T \sim B(10, p)$	M1	3.3
	$p = P(S \geq 15) = 0.299126\dots$	A1	1.1b
	So $P(T \geq 5) = 0.1487\dots$ awrt <u>0.149</u>	A1	1.1b
		(3)	
(c)	n is large and p close to 0.5	B1	1.2
		(1)	
(d)	$X \sim N(132, 59.4)$	B1	3.4
	$P(X \geq 149.5) = P\left(Z \geq \frac{149.5 - 132}{\sqrt{59.4}}\right)$	M1	1.1b
	$= 0.01158\dots$ awrt <u>0.0116</u>	A1cso	1.1b
		(3)	
(e)	e.g The probability is very small therefore there is evidence that the company's claim is incorrect.	B1	2.2b
		(1)	
(9 marks)			

Notes:
(a) B1: cao
(b) M1: for selection of an appropriate model for T 1 st A1: for a correct value of the parameter p (accept 0.3 or better) 2 nd A1: for awrt 0.149
(c) B1: both correct conditions
(d) B1: for correct normal distribution M1: for correct use of continuity correction A1: cso
(e) B1: correct statement

Q2.

Question	Scheme	Marks	AOs
(a)	$H_0: p = 0.1 \quad H_1: p \neq 0.1$	B1	2.5
		(1)	
(b)	Use of $X \sim B(50, 0.1)$ implied by sight of one of awrt 0.0052 or awrt 0.9755 or awrt 0.0245	M1	3.4
	Critical regions $X = 0$ or $X \geq 10$	A1	1.1b
	$X = 0$ and $X \geq 10$ plus $P(X = 0) = \text{awrt } 0.0052$ and $P(X \geq 10) = \text{awrt } 0.0245$	A1	1.1b
	SC: Both CR correct with no probabilities and no distribution seen scores M0A1A0		
(c)		(3)	
	0.0297	B1ft	1.1b
		(1)	
(d)	15 is <u>in the critical region</u> therefore there is evidence to support the <u>manager's</u> belief	B1ft	2.2b
		(1)	
(6 marks)			

Notes			
(a)	B1	For both hypotheses in terms of p or π . Connected to H_0 and H_1 correctly Condone 10% but not 10	
(b)	M1	Using correct distribution to find the probability associated with one tail of the CR If the correct distribution is <u>stated</u> (may be seen in part(a)) allow for one tail of the correct CR or one of (awrt 0.025 or awrt 0.005 or awrt 0.975) seen connected to a correct probability statement	
	A1	Lower CR $X = 0 / X < 1 / X \leq 0 /$ [condone eg $P(X = 0)$ labelled as CR] Or Upper CR $X \geq 10$ or $X > 9$ [condone $P(X \geq 10)$ or labelled as CR]	
	A1	Both CR's correct with the relevant probabilities Allow $\cup$ for "and" and $X > 9, X < 1, X \leq 0$ [do not allow $P(X = 0)$ or $P(X \geq 10)$ or] Allow CR in different form eg $(9, \infty), [10, \infty)$	
(c)	B1ft	awrt 0.0297 or 2.97% or ft for the sum of the probabilities in (b) for "their 2 critical regions" if seen. If none seen it must be awrt 0.0297 SC M0 in (b) for a one tail test Allow B1ft for their one tail CR in (b) eg 0.0338 or 0.0245 or 0.0579	
(d)	B1ft	A correct statement about 15 and "their CR" or sight $P(X \geq 15) = 0.0000738...$ and comparison with "their 0.0245" and a compatible correct statement in context. eg There is evidence that there has been a change in the <u>proportion/probability</u> arriving <u>late</u> Condone increase rather than change Do not allow contradicting statements. NB No CR given in (b) then B0	